

Certified Private Quantum Time Ticks Away Faster than Any Classical Clock

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Abstract: We introduce the concept of an entangled clock, where the flow of time is operationally defined by the discrete registration of measurement outcomes on a singlet state. Comparing the synchronization rate of two such clocks against classical models, we find that at obtuse relative angles, the quantum clock ticks strictly faster than this classical benchmark. We further propose a protocol for Certified Private Time, adapting device-independent randomness certification to the temporal domain; this guarantees a sovereign timeline that ticks away faster than any local realistic mechanism allows.

Keywords: quantum clock; entanglement; synchronization; Bell inequality; private time; quantum metrology

1. Introduction

The question of the fundamental nature of time and its measurement has recently been revisited through two distinct but complementary lenses: operational metrology in relativity and information-theoretic constraints in quantum foundations.

On the metrological side, the “dialogue” regarding the number of fundamental constants [1] established a consensus that dimensionless quantities are the ultimate observables. More recently, Matsas, Pleitez, Saa, and Vanzella [2] argued that a relativistic spacetime requires only a single fundamental dimensional constant. By employing “bona fide clocks,” one can define the spacetime metric entirely through proper time measurements, rendering spatial rulers redundant. This “single-unit universe”—a cosmos whose metric geometry is fixed by a single dimensional constant, namely proper time—relies on the mathematical rigidity of the Alexandrov-Zeeman theorems [3,4]. These theorems state that the causal structure (light cones) determines the geometry of Minkowski spacetime up to a global conformal factor. The physical clock breaks this dilation symmetry, fixing the scale of the universe [5].

If a single clock fixes the scale of the universe, a natural question arises: how do two such clocks, spatially separated but sharing a quantum history, measure the flow of time relative to one another? In classical relativity, synchronization is a matter of signaling and convention (Einstein synchronization [6]). In quantum mechanics, by contrast, synchronization can be mediated by entanglement, which introduces nonclassical correlations between distant systems. These correlations are constrained by no-go theorems such as Bell’s, and thus cannot in general be accounted for by local realistic statistics.

This brings us to the second lens: the study of quantum correlations. Zeilinger [7] proposed a foundational principle: that the irreducible randomness of quantum events arises because an elementary system carries only one bit of information. In an entangled state, this information is exhausted by the joint properties, leaving the individual constituents completely undefined. Simultaneously, Peres [8] demonstrated that classical correlations are

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strictly bounded because one can validly imagine the results of “unperformed experiments.” In contrast, quantum correlations defy these bounds precisely because such counterfactuals are illegitimate.

In this paper, we unify these perspectives to propose the concept of an *Entangled Clock*. We define the “tick” of a clock operationally as the registration of a measurement outcome. We show that the synchronization rate of two such clocks is not merely a function of relativistic geometry, but of the information-theoretic constraints on the system. By comparing the quantum prediction against Peres’ classical linear model, we identify a regime where the quantum clock “ticks faster” (shows higher coincidence rates) than this natural classical benchmark and, when correlations at several angles are taken into account, faster than any local realistic model allows.

2. The Entangled Clock Protocol

We construct a distributed time standard using a sequence of bipartite quantum systems. Let a source emit pairs of spin-1/2 particles (or polarization-entangled photons) in the singlet Bell state $|\psi^-\rangle$:

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}(|+\rangle_A |-\rangle_B - |-\rangle_A |+\rangle_B), \quad (1)$$

where A and B denote two spatially separated observers (Alice and Bob). Each observer possesses a measurement apparatus oriented along directions defined by unit vectors $\vec{a}$ and $\vec{b}$, respectively.

2.1. Operational Time and Zeilinger’s Principle

We define operational time T on each side not as a continuous background parameter, but as the accumulation of discrete events. Specifically, whenever a detector records a specific outcome (say, “up” or $+1$), the local clock increments by one unit:

$$T_i \rightarrow T_i + 1 \quad \text{if } \sigma_i = +1, \quad i \in \{A, B\}. \quad (2)$$

If the outcome is “down” (-1), the clock implies a “pause” or a non-tick for that interval.

According to Zeilinger’s Foundational Principle [7], an elementary system represents the truth value of a single proposition (1 bit). For a composite system of two spin-1/2 particles, the total information capacity is 2 bits. In the singlet state $|\psi^-\rangle$, these 2 bits are entirely exhausted by the joint propositions (e.g., “spins are opposite along z ” and “spins are opposite along x ”). Consequently, the system contains *zero* information about individual properties along any specific direction $\vec{n}$.

$$P(+)_\vec{n} = P(-)_\vec{n} = \frac{1}{2}. \quad (3)$$

Thus, the *local* proper time measured by Alice or Bob flows stochastically. Over a large ensemble of N pairs, the local clock will record approximately $N/2$ ticks. This randomness is not merely epistemic; it is ontologically necessary because the individual particle carries no answer to the question “are you up or down?” until the measurement is performed.

2.2. Synchronized Time

While local time is random, the relative time evolution between Alice and Bob is structured by entanglement. We define the *synchronized time rate* $R(\theta)$ as the probability that both clocks tick simultaneously in a given run (a coincidence count).

$$R(\theta) = P_{++}(\vec{a}, \vec{b}), \quad (4)$$

where θ is the angle between the measurement vectors $\vec{a}$ and $\vec{b}$. 74

For the singlet state, quantum mechanics predicts the joint expectation value $E_{QM}(\theta) = \langle \sigma_{\vec{a}} \otimes \sigma_{\vec{b}} \rangle = -\vec{a} \cdot \vec{b} = -\cos \theta$. The probability of a coincident tick $(+1, +1)$ is derived as follows: 75
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$$\begin{aligned} R_{QM}(\theta) &= \frac{1}{4}(1 + \langle \sigma_A \rangle + \langle \sigma_B \rangle + \langle \sigma_A \sigma_B \rangle) \\ &= \frac{1}{4}(1 + 0 + 0 - \cos \theta) \\ &= \frac{1}{2} \sin^2 \frac{\theta}{2}. \end{aligned} \quad (5)$$

Implementation Note: While we describe the protocol using spin-1/2 particles, an equivalent implementation uses polarization-entangled photon pairs. In that case, $|+\rangle / |-\rangle$ correspond to horizontal/vertical polarizations, and the detectors are linear polarizers. A “tick” corresponds to a photon passing through the polarizer and triggering an avalanche photodiode. 78
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3. Classical Simulation: Peres’ Bomb 83

To determine if the behavior of the entangled clock is unique, we must compare it to a classical standard. A robust classical analog for the singlet state was provided by Peres [8] and is often referred to as the “bomb fragment” model. 84
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Consider a bomb initially at rest (total angular momentum $\vec{J} = 0$) that explodes into two fragments with opposite angular momenta $\vec{J}_1$ and $\vec{J}_2 = -\vec{J}_1$. The direction of $\vec{J}_1$ is random and distributed uniformly over the sphere. The observers measure the sign of the projection of the angular momentum onto their axes $\vec{a}$ and $\vec{b}$: 87
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$$r_\alpha = \text{sign}(\vec{a} \cdot \vec{J}_1), \quad r_\beta = \text{sign}(\vec{b} \cdot \vec{J}_2). \quad (6)$$

Peres derives the correlation function for this macroscopic local hidden variable (LHV) model geometrically. The correlation depends on the overlap of the hemispheres defined by the vectors $\vec{a}$ and $\vec{b}$ on the unit sphere of possible angular momenta. The classical expectation value is strictly linear in θ : 91
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$$E_{cl}(\theta) = -1 + \frac{2\theta}{\pi} \quad \text{for } \theta \in [0, \pi]. \quad (7)$$

The corresponding classical synchronized tick rate is: 95

$$R_{cl}(\theta) = \frac{1}{4}(1 + E_{cl}) = \frac{1}{4} \left(1 - 1 + \frac{2\theta}{\pi} \right) = \frac{\theta}{2\pi}. \quad (8)$$

4. Comparison: When Quantum Clocks Tick Faster 96

In what follows, “ticking faster” will refer specifically to the *synchronized* tick rate $R(\theta) = P_{++}(\vec{a}, \vec{b})$ of the two clocks, i.e., to the density of joint $+1$ events per emitted pair. The local tick rates on Alice’s and Bob’s sides remain the same in both the quantum and classical models. 97
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We now compare the two synchronization rates. We can distinguish three “cardinal” regimes where the physics appears identical, and one “anomalous” regime where the quantum clock diverges. 101
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4.1. The Cardinal Regimes

1. **Perfect Anti-Synchrony** ($\theta = 0$): Detectors are aligned ($\vec{a} = \vec{b}$).

$$R_{QM}(0) = 0, \quad R_{cl}(0) = 0.$$

Because $|\psi^-\rangle$ is perfectly anti-correlated (singlet), if Alice's clock ticks, Bob's does not. The clocks are perfectly out of phase.

2. **Non-Correlation** ($\theta = \pi/2$): Detectors are orthogonal ($\vec{a} \perp \vec{b}$).

$$R_{QM}(\pi/2) = 1/4, \quad R_{cl}(\pi/2) = 1/4.$$

Here, $E_{QM} = 0$ and $E_{cl} = 0$. The clocks are statistically independent.

3. **Perfect Synchrony** ($\theta = \pi$): Detectors are anti-aligned ($\vec{a} = -\vec{b}$).

$$R_{QM}(\pi) = 1/2, \quad R_{cl}(\pi) = 1/2.$$

Since $|\psi^-\rangle$ implies opposite spins, measuring opposite directions yields identical results ($++$ or $--$). The clocks tick together 50% of the time.

In these regimes, the informational constraint of the quantum bit yields the same statistics as the classical angular momentum conservation.

4.2. The Anomalous Regime

The divergence between the single-unit universe governed by quantum information and one governed by classical statistical mechanics appears at intermediate angles. We define the *synchronization excess* $\Delta(\theta)$ as:

$$\Delta(\theta) \equiv R_{QM}(\theta) - R_{cl}(\theta) = \frac{1}{2} \sin^2 \frac{\theta}{2} - \frac{\theta}{2\pi}. \quad (9)$$

To find the maximum deviation, we find the extrema by setting $d\Delta/d\theta = 0$:

$$\frac{1}{4} \sin \theta - \frac{1}{2\pi} = 0 \implies \sin \theta = \frac{2}{\pi}. \quad (10)$$

This yields two solutions in $[0, \pi]$:

$$\theta_1 = \arcsin(2/\pi) \approx 0.69 \text{ rad } (39.5^\circ), \quad (\Delta < 0), \quad (11)$$

$$\theta_2 = \pi - \theta_1 \approx 2.45 \text{ rad } (140.5^\circ), \quad (\Delta > 0). \quad (12)$$

At $\theta_1 \approx 39.5^\circ$, the classical correlation is stronger (more negative), so the quantum clock ticks *less* frequently than the classical prediction. This is the regime where the classical linear correlation drops faster than the quantum cosine.

However, at $\theta_2 \approx 140.5^\circ$, the quantum clock exhibits a significant “speedup” (see Fig. 1).

- Classical Rate: $R_{cl}(140.5^\circ) \approx 0.390$.
- Quantum Rate: $R_{QM}(140.5^\circ) \approx 0.443$.

In this configuration, the quantum clocks tick in unison roughly 13.6% more often than their classical counterparts.

The cosine modulation of the quantum correlation function, which is consistent with Zeilinger's information-invariance principle, permits a higher density of coincident events than the linear constraint imposed by the isotropic bomb-fragment model. In particular, at θ_2 the quantum prediction exceeds the linear classical benchmark by about 13.6%. Figure 2 explicitly shows this synchronization excess $\Delta(\theta)$. The positive lobe represents a regime

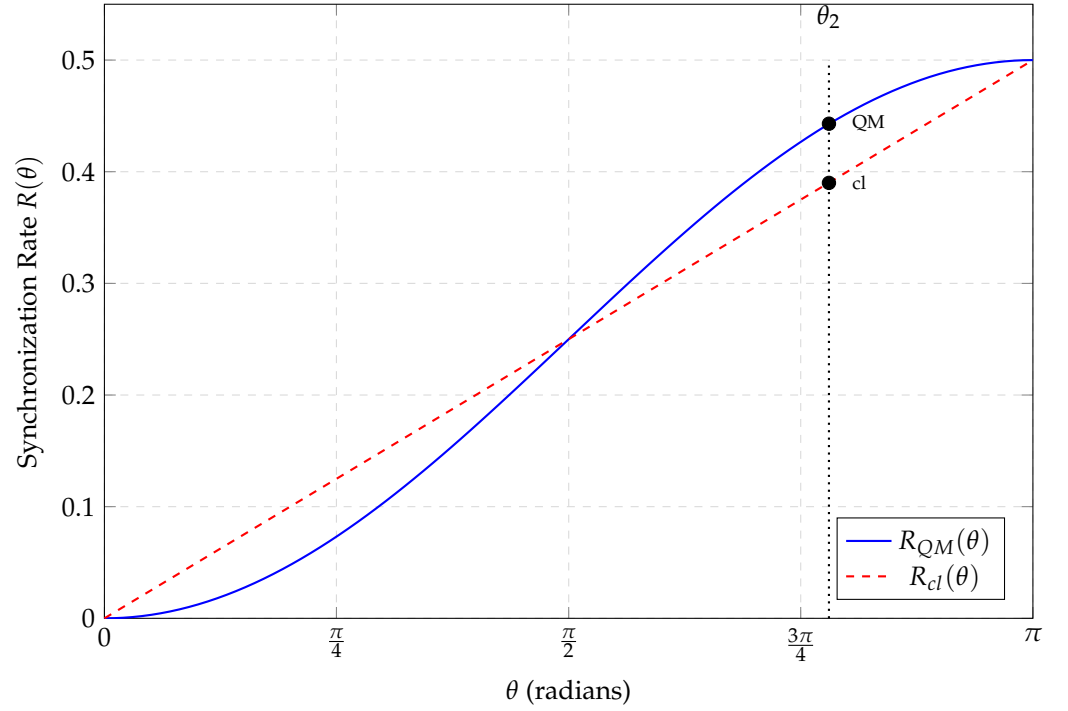


Figure 1. Comparison of the quantum (R_{QM} , solid blue) and classical (R_{cl} , dashed red) synchronization rates as functions of the relative detector angle θ . The curves coincide at $\theta = 0, \pi/2, \pi$ but diverge maximally at $\theta_2 \approx 140.5^\circ$, where the quantum clock “ticks faster.”

where, if one demands a single local realistic model that applies to *all* angles simultaneously, the quantum predictions force the clocks to agree more often than such a model can allow.

5. Contextuality and Unperformed Experiments

Why does the quantum clock tick faster? The answer lies in the nature of the information carried by the system and the role of contextuality.

In Peres’ model [8], the angular momentum vector $\vec{J}$ exists independently of the measurement. Peres argues that Bell’s inequality can be derived by constructing a table of outcomes for experiments actually performed (settings $\vec{a}, \vec{b}$) and hypothetical experiments *not* performed (settings $\vec{a}', \vec{b}'$). For a classical system, this table must be internally consistent because the value $\vec{J}$ pre-exists and determines the outcome for any possible question. In the specific bomb-fragment model, the overlap of the corresponding hemispheres on the unit sphere yields a strictly linear correlation function, Eq. (7), which in turn strictly limits the synchronized tick rate.

In the quantum case, following Zeilinger [7], the bipartite system carries only 2 bits of information, which are already exhausted by the specific joint state. There is no “spare” information to encode definite answers to all possible, mutually incompatible measurement questions along directions $\vec{a}', \vec{b}'$. From this information-theoretic perspective, the system is not constrained to maintain consistency across a full table of actual and counterfactual outcomes, and thus is not subject to the Boole–Bell-type constraints [9,10] that enforce the existence of a single joint probability distribution for all settings and, in particular, lead to linear behavior in models like Peres’ bomb. This “freedom from counterfactual definiteness” is one way to understand why the quantum correlation $E_{QM}(\theta)$ follows the cosine law and exceeds the linear bomb-model bound at $\theta \approx 140^\circ$.

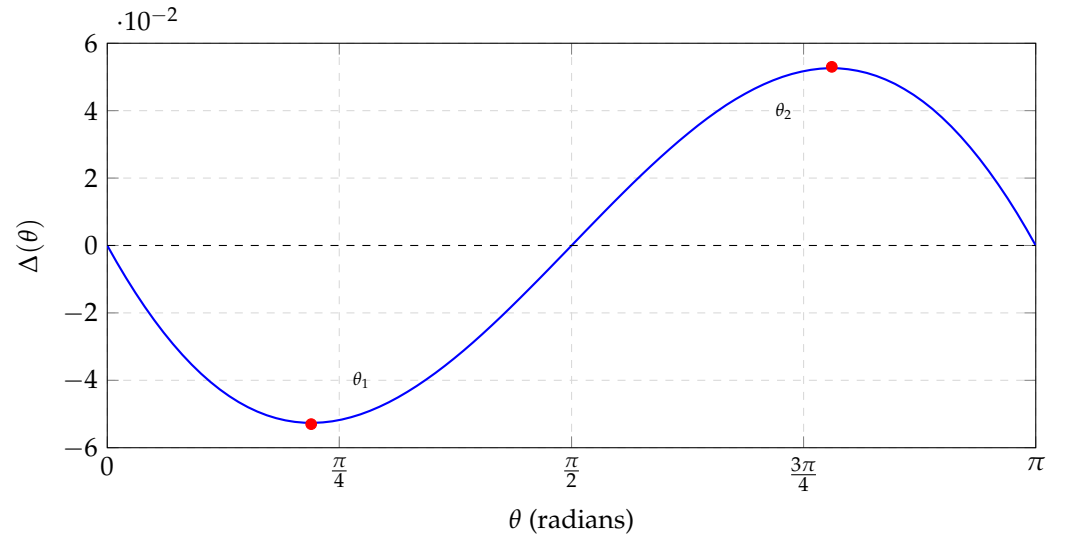


Figure 2. The synchronization excess $\Delta(\theta) = R_{QM} - R_{cl}$. The quantum clock lags at acute angles ($\theta_1 \approx 39.5^\circ$) but leads at obtuse angles ($\theta_2 \approx 140.5^\circ$).

5.1. Certifying Quantumness

While the rate $R_{QM} \approx 0.44$ is notably higher than the standard classical rate $R_{cl} \approx 0.39$, one must be cautious. A single measurement setting cannot distinguish quantum from classical clocks. As noted in recent work on contextuality [11], for any *single* fixed angle θ^* , one can construct a local classical model (e.g., Aerts’ “broken elastic band” model [12]) that reproduces the quantum probability $R_{QM}(\theta^*)$ exactly. In such models, the hidden variable space is deformed or the probability density is non-uniform to mimic the cosine statistics locally.

The signature of quantum synchronization is *contextual*: the same underlying state produces the *entire* cosine curve across all angles simultaneously. To certify quantumness, Alice and Bob must therefore vary their angles and compare several correlations at once. Any local realistic model must admit a single joint probability distribution that reproduces all these correlations and thus obeys Bell-type inequalities, whereas the quantum clock yields a family of cosine values that violates these bounds. The enhanced synchronized tick rate in the obtuse regime is then seen not as an isolated anomaly, but as one facet of a globally nonclassical correlation structure.

The excess synchronization rate $\Delta(\theta) > 0$ in the obtuse regime is directly related to violations of Bell-type inequalities such as the Clauser–Horne–Shimony–Holt (CHSH) inequality [13]. The CHSH parameter S is composed of four correlation terms:

$$S = |E(\vec{a}, \vec{b}) - E(\vec{a}, \vec{b}') + E(\vec{a}', \vec{b}) + E(\vec{a}', \vec{b}')|. \quad (13)$$

The classical bound $S \leq 2$, obtained from the convex hull computation of the corresponding correlation polytope [9,10], arises from the requirement that all four correlations be marginals of a single joint probability distribution, that is, from the assumption that the table of actual and counterfactual outcomes is internally consistent. The linear correlation function (7) is one explicit realization of such a local realistic model. The quantum violation (approaching the Tsirelson bound $2\sqrt{2}$) and the enhanced synchronization rate in the obtuse regime are two manifestations of the same underlying phenomenon.

6. Experimental Considerations

Implementing this protocol requires addressing practical constraints. In an ideal scenario, every particle pair results in a potential tick. In reality, photon detectors have finite efficiency $\eta < 1$.

If $\eta < 1$, the clocks will frequently “miss” ticks even when the quantum outcome would have been $+1$. However, the synchronization rate $R(\theta)$ is typically measured as a coincidence rate normalized by the total number of emitted pairs (or inferred from singles counts). If we define the measured rate $R_{exp}(\theta)$, we have:

$$R_{exp}(\theta) \approx \eta_A \eta_B R_{QM}(\theta). \quad (14)$$

This reduced efficiency lowers the absolute rate of ticking for both classical and quantum models. The crucial comparison is the *relative* rate or the visibility of the interference fringe. A high-visibility experiment (using, for example, spontaneous parametric down-conversion and superconducting nanowire single-photon detectors with $\eta > 0.9$) would clearly resolve the difference between the linear classical prediction and the sinusoidal quantum prediction.

Furthermore, the definition of “simultaneity” requires a coincidence window τ_c . If the path lengths to Alice and Bob differ, or if there is electronic jitter, ticks might not register as simultaneous. Standard quantum optical techniques allow τ_c to be on the order of nanoseconds. The “Entangled Clock” is thus a statistical construct: it does not provide a continuous readout for a single run but establishes a time scale over an ensemble of events, where the density of synchronized events serves as the metrological standard.

7. Certified Private Time

The discussion above highlights a crucial distinction: while a classical model can be engineered to reproduce quantum statistics for a single context, it fails to reproduce the full correlation structure across multiple contexts. This brings us to a stronger application of the entangled clock: the certification of “private time.”

As reviewed by Acín and Masanes [14], the violation of Bell inequalities serves as a device-independent certificate for randomness. In the context of random number generation (QRNG), this certification ensures that the output bits were not pre-determined by any hidden variable λ available to an adversary (Eve). Transposing this logic to metrology, we can conceive of a protocol for *Certified Private Time*.

7.1. The Memory-Stick Clock

Consider the temporal equivalent of the “memory-stick attack” described in Ref. [14]. An adversary, Eve, manufactures two clocks and gives them to Alice and Bob. These clocks are classical devices containing a pre-recorded sequence of ticks derived from a hidden variable λ . To Alice and Bob, the ticks appear stochastic (e.g., mimicking radioactive decay), and they may even exhibit correlations. However, because the sequence is pre-determined, Eve possesses a “lookup table” for time. She knows exactly when Alice’s clock will tick before it happens. In such a scenario, Alice’s time is not her own; it is fully correlated with the adversary’s timeline.

In a classical universe (or one governed by local hidden variables), one cannot prove that a clock is not a playback device. The flow of time could be merely the reading of a script written at the Big Bang (super-determinism aside).

7.2. Protocol for Time Certification

The entangled clock allows Alice and Bob to rule out this “playback” scenario. By switching between measurement settings (angles $\vec{a}, \vec{a}'$ for Alice and $\vec{b}, \vec{b}'$ for Bob) and

accumulating the statistics of their coincident ticks, they can estimate the CHSH parameter S .

If they observe $S > 2$, they certify that the joint state does not admit a local realistic description. Physically, this implies that the specific outcomes (the ticks) did not exist prior to the measurement events. Consequently, the time signal generated by the clock is:

1. **Intrinsic:** The ticks are generated *in situ* by the act of measurement, not read from a memory.
2. **Private:** Because the outcomes are fundamentally indeterministic, they are uncorrelated with any external system, including Eve's variables.

7.3. Escaping the Block Universe

This leads to a radical reinterpretation of the “flow” of time. In a standard operational view, time is a parameter inferred from correlations between a clock system and the rest of the universe. However, if the clock's ticks are certified random via Bell violation, the clock is dynamically decoupled from the rest of the universe until the moment of the tick.

A “Certified Private Clock” thus ticks at a rate that is not only statistically distinct (the “faster” rate at obtuse angles discussed in Sec. IV) but ontologically distinct. It generates a local history that no external observer could have predicted. Just as device-independent protocols use Bell violation to expand a short private random seed into a distinct, private key, the entangled clock expands a short interval of private operational agency (the choice of settings) into a sustained, private flow of time.

This certification is robust against the “middleman attack” often cited in foundational debates. Even if Eve builds the source and controls the channels, she cannot force the devices to violate the CHSH inequality with pre-programmed ticks unless she knows the measurement settings (the context) in advance. To prevent this, Alice and Bob must ensure “measurement (parameter)independence” [15]. This can be achieved either through genuine free choice or, alternatively, by consuming a short, predefined private random key to govern the settings. In the latter case, the system functions as a temporal implementation of Acín's expansion protocol [14], growing the initial finite seed into a sustained, sovereign timeline that is mathematically guaranteed to be uncorrelated with any master clock Eve might hold.

8. Conclusion

We have proposed an operational protocol for *Certified Private Time*, in which spatially separated quantum clocks are synchronized via singlet-state correlations. By adapting device-independent certification methods [14], we showed that if observers consume a private seed to ensure measurement independence, the resulting timeline is certified to be “sovereign”—generated *in situ* and mathematically uncorrelated with any pre-existing external script.

Crucially, these quantum clocks differ from classical counterparts not just in their security, but in their rate. For instance, at obtuse relative angles ($\theta \approx 140^\circ$), the synchronized tick rate exhibits a “temporal speedup” of $\sim 13\%$ relative to Peres' classical bomb-fragment model. This enhancement obeys the conservation of probability: the quantum correlation function effectively “squeezes” the statistics, populating the synchronized states ($++$ and $--$) more densely than the linear classical bound allows, at the necessary expense of the mixed outcomes ($+-$ and $-+$).

This deviation is a direct signature of contextuality. Unlike the classical model, the quantum system is not constrained to maintain consistency with unperformed measurements. The violation of Bell-type inequalities thus serves a dual purpose: it certifies the privacy of the time stream against a middleman adversary, and it confirms that the

sovereign quantum clock ticks away faster than allowed by the constraints of classical reality.

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